

Healthcare Synthetic Data Data Dictionary

Version 2.0 — 103 columns

Five-domain integrated healthcare encounter dataset

Clinical · Operational · Financial · Patient Experience · HR · Supply Chain
FHIR R4 · ICD-10-CM · CPT · LOINC · RxNorm · CARC · HIPAA Safe Harbor
HEDIS 2025 · CAHPS · ISO 7101 · HIMSS EMRAM · HACRP · BPCI

Lead Healthcare Data Architect — SDG Engine v2.0

April 2026

Introduction

This document describes every column in the synthetic healthcare encounter dataset produced by the SDG Engine v2.0. The dataset contains 103 columns across eight logical sections covering five tightly linked domains and is designed to support analytics, machine learning, and quality improvement workflows that span the full hospital management lifecycle.

What's new in version 2.0

Version 2.0 adds 11 new KPI columns to support additional dashboard requirements identified during project review. New columns are highlighted in light orange throughout the document and tagged with [NEW] in the field-of-interest column. The new fields are distributed across four domains:

- Clinical (2 fields): hospital-acquired infection tracking and post-discharge follow-up compliance.
- Operational (1 field): physician assessment timestamp for the door-to-doctor KPI.
- Financial (5 fields): claim denial flag, denial reason code, equipment type and usage minutes, and consumable supply cost.
- Patient experience (3 fields): telehealth adoption, appointment scheduling lead time, and patient-reported wait time.

Purpose

This data dictionary serves three audiences:

- Data scientists and analysts who need to understand each field before modeling.
- Clinical informaticists who must validate that the synthetic data correctly mirrors real-world clinical and operational patterns.
- Compliance and governance teams who need to verify regulatory alignment with HIPAA, HEDIS, ISO 7101, and HIMSS standards.

Domain coverage

Each row in the dataset represents a single inpatient surgical encounter and contains fields drawn from eight logical sections:

1. Patient demographics — six de-identified attributes describing the patient.
2. Clinical — twenty-five fields covering diagnosis, vitals, labs, medications, outcomes, HAI, and follow-up.
3. Operational — eighteen fields covering patient flow, physician assessment, OR utilization, and safety events.
4. Financial — fifteen fields covering claims, charges, denials, FWA detection, equipment, and consumables.
5. Human resources — twenty-one fields covering staffing, burnout, and CAHPS patient experience.
6. Patient experience and engagement — three fields covering telehealth, scheduling lead time, and perceived wait.
7. Supply chain — eleven fields covering surgical kits, inventory, and demand forecasting.

8. Quality and standards — four fields capturing HEDIS, ISO 7101 PDSA, and HIMSS EMRAM benchmarking.

Cross-domain correlations

Unlike a random data generator, the SDG Engine enforces statistically valid relationships between domains. The most important linkages are summarized below; readers should consult the per-field descriptions for the precise mechanism in each case.

- Severity index drives length of stay, OR time, charges, and 30-day readmission probability.
- Comorbidities drive specific lab values: diabetes elevates HbA1c and glucose; CHF elevates NT-proBNP.
- Patient-to-nurse ratio combined with nurse burnout increases the probability of safety events.
- Burnout and patient load are inversely correlated with CAHPS patient experience scores.
- Procedure type drives surgical kit selection, which in turn drives supply chain demand and cost.
- Fraud upcoding flags inflate the submitted charge to create a learnable signal for FWA models.
- Length of stay, ICU admission, and severity drive hospital-acquired infection probability. [NEW]
- Fraud indicators dramatically increase claim denial probability and dictate the denial reason code. [NEW]
- Patient age inversely correlates with telehealth adoption (digital divide effect). [NEW]
- Bed occupancy delays physician assessment, modeling capacity strain on door-to-doctor times. [NEW]

Regulatory compliance

All fields have been generated in accordance with the HIPAA Safe Harbor de-identification method (45 CFR §164.514(b)(2)). The 18 direct identifiers have been removed or generalized: no names, no specific dates other than year, no geographic units smaller than a state, no identifying numbers (SSN, MRN, account numbers, license numbers, device identifiers), and no biometric or facial data. Patient and provider identifiers in this dataset are synthetic tokens with no link to any real person.

Clinical terminologies follow the FHIR R4 specification and use ICD-10-CM (FY2025) for diagnoses, CPT and HCPCS for procedures, LOINC for laboratory observations, RxNorm for medications, and X12 CARC codes for claim adjustment reason codes. Quality benchmarks align with HEDIS 2025 measure specifications, the HCAHPS survey instrument, ISO 7101 healthcare quality management requirements, the HIMSS EMRAM digital maturity framework, the CMS Hospital-Acquired Condition Reduction Program (HACRP), and the CMS Bundled Payments for Care Improvement (BPCI) program.

1. Patient demographics (HIPAA Safe Harbor compliant)

Six de-identified attributes describing the patient population. All 18 HIPAA direct identifiers have been removed.

Column name	Type	Units / format	Field of interest	Description and purpose
patient_id	String	PAT-XXXXXX	Identifier	Synthetic patient identifier with no link to any real person. Used as the primary key for joining all clinical, financial, and operational records belonging to the same individual. Format is a sequential token, not a real medical record number.
birth_year	Integer	YYYY	Demographics	Year of birth only. Per HIPAA Safe Harbor §164.514(b)(2)(i)(C), specific dates other than year are considered direct identifiers and have been suppressed. Year alone is permitted for patients under 90 and supports age-based cohort analysis.
age	Integer	Years	Demographics	Computed age at the time of the encounter. Range capped at 95 to comply with the HIPAA Safe Harbor rule that ages over 89 must be aggregated into a single category. Used for age-stratified clinical analysis and risk modeling.
sex	String	male / female	Demographics	Administrative sex at the time of registration. Used for clinical decision support, sex-specific reference ranges (e.g., hemoglobin), and equity analysis.
race_ethnicity	String	Category	Demographics / Equity	Self-reported race and ethnicity category (White, Black, Hispanic, Asian, Other). Distribution approximates U.S. census proportions. Used for health equity analysis, disparities research, and CMS stratified quality reporting.
state	String	USPS 2-letter	Demographics	U.S. state of residence at state level only. Per HIPAA Safe Harbor, geographic subdivisions smaller than a state (ZIP, county) are suppressed when the geographic unit contains fewer than 20,000 people. Used for regional analysis and care network planning.

2. Clinical domain — diagnosis, vitals, labs, and outcomes

Twenty-five fields capturing the patient's clinical state at the time of encounter, including ICD-10 diagnoses, vital signs, LOINC-coded lab results, medication adherence, the 30-day readmission outcome, hospital-acquired infection tracking, and post-discharge follow-up compliance.

Column name	Type	Units / format	Field of interest	Description and purpose
<code>encounter_id</code>	String	UUID v4	Identifier	Globally unique identifier for the encounter. Maps to the FHIR Encounter.id resource. Acts as the foreign key linking clinical, operational, financial, HR, and supply chain records for a single hospital visit.
<code>icd10_code</code>	String	ICD-10-CM	Clinical / Diagnosis	Primary diagnosis code from the ICD-10-CM terminology (FY2025). Drives DRG assignment, risk-adjustment, and HEDIS denominator inclusion. Examples: M17.11 (right knee osteoarthritis), I25.10 (atherosclerotic heart disease), C34.90 (lung cancer).
<code>diagnosis_display</code>	String	Free text	Clinical / Diagnosis	Human-readable description of the ICD-10 code, drawn directly from the official CMS code descriptor list. Used for reporting and clinician-facing displays.
<code>diagnosis_category</code>	String	Category code	Clinical / Service Line	High-level service line grouping for the diagnosis (MSK, GI, CV, ONC, ENDO, GU, PULM, DERM). Used for service-line dashboards, surgeon assignment, and supply chain forecasting.
<code>severity_index</code>	Float	0.0–1.0	Clinical / Risk	Synthetic acuity score that drives downstream correlations across all five domains. Higher values increase length of stay, OR time, charges, and readmission probability. Designed to mimic the role that APR-DRG severity-of-illness or Charlson Comorbidity Index plays in real-world risk adjustment.
<code>comorbidity_count</code>	Integer	Count	Clinical / Risk	Number of chronic comorbid conditions present (diabetes, hypertension, CHF). Used as a simplified Charlson-style index for risk stratification and readmission modeling.

Column name	Type	Units / format	Field of interest	Description and purpose
has_diabetes	Binary	0 / 1	Clinical / Chronic Condition	Indicates Type 2 diabetes mellitus comorbidity (ICD-10 E11.9). Triggers HEDIS 2025 inclusion for the Comprehensive Diabetes Care measure and drives elevated HbA1c, glucose, and creatinine values.
has_hypertension	Binary	0 / 1	Clinical / Chronic Condition	Indicates essential hypertension comorbidity (ICD-10 I10). Drives elevated systolic and diastolic blood pressure values and triggers Lisinopril prescribing in the medication record.
has_chf	Binary	0 / 1	Clinical / Chronic Condition	Indicates congestive heart failure comorbidity (ICD-10 I50.9). Drives elevated NT-proBNP values and contributes to higher readmission probability.
sbp_mmhg	Float	mmHg	Clinical / Vital Sign	Systolic blood pressure measurement at presentation. Generated from a Gaussian distribution with mean shifted upward for hypertensive patients. Normal range: 90–120 mmHg.
dbp_mmhg	Float	mmHg	Clinical / Vital Sign	Diastolic blood pressure at presentation. Correlated with systolic value and hypertension status. Normal range: 60–80 mmHg.
heart_rate_bpm	Float	Beats / minute	Clinical / Vital Sign	Heart rate at presentation. Mean shifted upward by severity to simulate physiological response to acute illness. Normal resting range: 60–100 bpm.
spo2_pct	Float	Percent	Clinical / Vital Sign	Peripheral capillary oxygen saturation. Inversely correlated with severity to reflect respiratory compromise in higher-acuity cases. Normal range: 95–100%.
temperature_f	Float	Degrees Fahrenheit	Clinical / Vital Sign	Body temperature at presentation. Slight elevation correlates with severity to model infection or inflammatory response. Normal: 97.0–99.0°F.
bmi	Float	kg/m ²	Clinical / Anthropometric	Body mass index. Mean elevated for diabetic patients to reflect real-world correlation between obesity and Type 2

Column name	Type	Units / format	Field of interest	Description and purpose
				diabetes. WHO categories: <18.5 underweight, 18.5–24.9 normal, 25–29.9 overweight, ≥30 obese.
respiratory_rate	Integer	Breaths / minute	Clinical / Vital Sign	Respiratory rate at presentation. Mean increases with severity to reflect respiratory distress. Normal adult range: 12–20 breaths/minute.
hba1c_pct	Float	Percent	Clinical / Lab (LOINC 4548-4)	Hemoglobin A1c — a 90-day average of blood glucose. The cornerstone HEDIS 2025 measure for diabetic patients. Diabetic patients receive elevated values (mean 8.2%); non-diabetics receive normal values (mean 5.2%). Values >9% trigger the HEDIS poor-control care gap.
glucose_mg_dl	Float	mg/dL	Clinical / Lab (LOINC 2345-7)	Serum glucose level. Normal fasting: 70–100 mg/dL. Elevated in diabetic and high-severity patients. Used for acute glycemic management.
creatinine_mg_dl	Float	mg/dL	Clinical / Lab (LOINC 2160-0)	Serum creatinine — a marker of kidney function. Elevated values in higher-severity patients reflect acute kidney injury risk. Normal: 0.6–1.2 mg/dL.
wbc_10e3_u1	Float	10 ³ cells/μL	Clinical / Lab (LOINC 6690-2)	White blood cell count. Elevated in higher-severity patients to reflect infection or inflammatory response. Normal: 4.5–11.0 ×10 ³ /μL.
nt_probnp_pg_ml	Float	pg/mL	Clinical / Lab (LOINC 33762-6)	N-terminal pro-B-type natriuretic peptide — a biomarker of cardiac strain. Dramatically elevated in CHF patients (mean ~1,800 pg/mL) versus normal (<125 pg/mL). Used for heart failure diagnosis and risk stratification.
medication_adherence_pdc	Float	0.0–1.0	Clinical / Pharmacy	Proportion of Days Covered — the standard CMS Star Rating measure for medication adherence. PDC ≥0.80 is considered adherent. Inversely correlated with severity to model the real-world challenge of polypharmacy in sicker patients.

Column name	Type	Units / format	Field of interest	Description and purpose
readmission_30d_flag	Binary	0 / 1	Clinical / Outcome	Indicates whether the patient was readmitted within 30 days of discharge. Probability is computed from severity, age, comorbidity count, and medication adherence. Aligns with the CMS Hospital Readmissions Reduction Program (HRRP) target measure.
hai_flag	Binary	0 / 1	Clinical / Quality and Safety [NEW]	Hospital-Acquired Infection flag. Indicates whether the patient developed an infection during their inpatient stay (CAUTI, CLABSI, SSI, C. difficile, ventilator-associated pneumonia). Baseline rate ~3%, increased by long length of stay (+1.5% per day beyond 3), ICU admission (+4%), oncology service (+3%), and severity (up to +4%). The numerator for the CMS Hospital-Acquired Condition Reduction Program (HACRP) measure and a key driver of CMS value-based purchasing penalties.
follow_up_within_48h_flag	Binary	0 / 1	Clinical / Care Transition [NEW]	Indicates whether the patient received documented follow-up care (clinic visit, phone call, or telehealth contact) within 48 hours of discharge. Baseline rate ~65%, increased by severity (sicker patients receive more vigilant care management) and CAHPS care transition scores. A leading indicator for 30-day readmission prevention and a core measure in the CMS Bundled Payments for Care Improvement (BPCI) program.

3. Operational domain — patient flow, OR utilization, and safety

Eighteen fields capturing the operational lifecycle of the encounter from triage through discharge, including OR scheduling, length of stay, bed assignment, physician assessment timing, and safety event tracking.

Column name	Type	Units / format	Field of interest	Description and purpose
triage_timestamp	DateTime	ISO 8601	Operations / Patient Flow	Time the patient was triaged in the emergency department or pre-admission unit. Marks the start of the patient's journey through the hospital. Used to compute door-to-doctor and door-to-bed metrics.
physician_assessment_timestamp	DateTime	ISO 8601	Operations / ER Throughput [NEW]	Time the patient was first evaluated by a physician (door-to-doctor moment). Door-to-doctor interval averages 30 minutes after triage and is delayed when bed occupancy exceeds 80%, reflecting the way capacity strain slows initial physician evaluation. Subtracting triage_timestamp from this field yields the door-to-doctor KPI, a core ED throughput measure tracked by the Emergency Department Benchmarking Alliance (EDBA) and CMS.
admit_timestamp	DateTime	ISO 8601	Operations / Patient Flow	Time the patient was formally admitted to the hospital. Marks the beginning of the inpatient encounter for length-of-stay calculations and bed-day accounting.
bed_request_time	DateTime	ISO 8601	Operations / Bed Management	Time at which a request for an inpatient bed was placed in the bed management system. Used to compute bed wait time and identify capacity bottlenecks.
bed_assign_time	DateTime	ISO 8601	Operations / Bed Management	Time at which a specific bed was assigned to the patient. The interval between bed_request_time and this field is the bed wait

Column name	Type	Units / format	Field of interest	Description and purpose
				time, a key throughput metric.
unit_assigned	String	Unit name	Operations / Bed Management	The inpatient unit to which the patient was assigned (ICU, MICU, SICU, Step-Down, Med-Surg, Ortho, Cardiac, Oncology, General). Drives nurse-staffing model, equipment availability, and cost center allocation.
bed_occupancy_pct	Float	Percent	Operations / Capacity	Unit-level bed occupancy at the time of assignment. Above 85% is generally considered the threshold at which patient throughput, safety, and staff stress begin to deteriorate. Used in capacity planning and surge modeling.
cpt_code	String	CPT/HCPCS	Operations / Procedure	Current Procedural Terminology code for the surgical procedure performed. Drives professional fee billing, OR scheduling, and surgical kit selection. Examples: 27447 (total knee arthroplasty), 47562 (laparoscopic cholecystectomy).
procedure_display	String	Free text	Operations / Procedure	Human-readable description of the CPT code from the AMA CPT terminology. Used for OR schedules, consent forms, and clinician-facing displays.
or_start	DateTime	ISO 8601	Operations / OR Utilization	Time the patient entered the operating room. Used for OR utilization analytics, surgeon punctuality measurement, and first-case-on-time-start metrics.
or_end	DateTime	ISO 8601	Operations / OR Utilization	Time the patient exited the operating room. Combined with or_start, this enables case-length analysis and surgeon-specific time benchmarking.
actual_or_minutes	Integer	Minutes	Operations / OR Utilization	Actual operating room duration. Generated as a Gaussian distribution

Column name	Type	Units / format	Field of interest	Description and purpose
				around the procedure-specific baseline OR time, with realistic variance to reflect case complexity. Used to identify outlier procedures and inform block-time scheduling.
or_turnover_minutes	Integer	Minutes	Operations / OR Utilization	Time between this case and the next case in the same operating room (room cleaning, setup, patient transfer). Industry benchmark target is typically <30 minutes. A key OR efficiency KPI.
los_days	Integer	Days	Operations / Throughput	Length of stay in days. Computed from severity, procedure complexity, and a stochastic component. Strongly correlated with severity index — the cornerstone metric for inpatient throughput, cost, and capacity planning.
discharge_timestamp	DateTime	ISO 8601	Operations / Patient Flow	Time the patient was discharged from the hospital. Marks the end of the inpatient encounter and the start of the 30-day readmission window.
safety_incident_flag	Binary	0 / 1	Operations / Quality and Safety	Indicates whether a patient safety event occurred during the encounter. Probability is driven by patient-to-nurse ratio, nurse burnout, and overtime hours, modeling the well-established link between staffing strain and safety events.
incident_type	String	Category	Operations / Quality and Safety	Type of safety event when one occurred (Fall, Medication Error, Pressure Injury, CAUTI, CLABSI, SSI, Near-Miss, Equipment Failure, Delay in Care). Categories align with NDNQI and AHRQ Common Formats.
incident_severity	String	Category	Operations / Quality and Safety	Severity classification of the safety event (Near-Miss, Minor, Moderate, Major). Used for root cause

Column name	Type	Units / format	Field of interest	Description and purpose
				analysis prioritization and for HAC (Hospital-Acquired Condition) reporting.

4. Financial domain — claims, charges, denials, equipment, and consumables

Fifteen fields capturing the financial side of the encounter, including the insurance claim, payer-specific allowed amounts, patient responsibility, fraud-waste-and-abuse (FWA) detection indicators, claim denial outcomes with denial reason codes, high-cost equipment utilization, and general consumable supply costs.

Column name	Type	Units / format	Field of interest	Description and purpose
<code>claim_id</code>	String	CLM-XXXXXXX	Financial / Identifier	Unique identifier for the insurance claim associated with this encounter. Maps to the FHIR Claim.id resource. Acts as the primary key for revenue cycle and FWA analytics.
<code>payer</code>	String	Payer name	Financial / Reimbursement	Primary insurance payer (Medicare, Medicaid, BlueCross, Aetna, UnitedHealth, Cigna, Self-Pay). Distribution mirrors a typical U.S. hospital payer mix. Drives the contractual discount applied to submitted charges.
<code>npi_billing</code>	String	10-digit NPI	Financial / Provider	National Provider Identifier of the billing surgeon. Synthetic, not linked to any real provider. Used for provider-level revenue analysis, productivity reporting, and physician compensation modeling.
<code>drg_weight</code>	Float	Relative weight	Financial / Reimbursement	Diagnosis Related Group relative weight, which determines Medicare and many commercial payments under prospective payment systems. Higher weights correspond to higher acuity and higher reimbursement. Computed from severity index plus stochastic variation.
<code>submitted_charge_usd</code>	Float	USD	Financial / Revenue Cycle	Total billed charges submitted on the claim, before payer adjudication. Computed from OR time, supply costs, facility per-diem, and a fraud-inflation factor when an upcoding flag is set. Represents the chargemaster value, not the actual collected amount.
<code>allowed_amount_usd</code>	Float	USD	Financial / Revenue Cycle	Payer-allowed amount after applying the contractual discount. This is the contractually permitted reimbursement,

Column name	Type	Units / format	Field of interest	Description and purpose
				regardless of how much the patient or insurer ultimately pays. Computed as $\text{submitted_charge} \times (1 - \text{payer_discount})$.
patient_responsibility_usd	Float	USD	Financial / Patient Liability	Portion of the allowed amount the patient is responsible for (deductibles, copays, coinsurance). For self-pay patients, equals the full allowed amount. Used for collections, financial counseling, and bad-debt forecasting.
fraud_upcoding_flag	Binary	0 / 1	Financial / FWA Detection	Indicates whether the claim shows signs of upcoding — billing for a higher level of service than was actually provided. Generated at a ~4% baseline rate. When flagged, the submitted charge is inflated by a factor of 1.4–2.2 to create a learnable signal for FWA models.
fraud_duplicate_flag	Binary	0 / 1	Financial / FWA Detection	Indicates a likely duplicate claim — the same service billed more than once. Generated at a ~1.5% baseline rate. Used for training duplicate-detection models.
fraud_unbundling_flag	Binary	0 / 1	Financial / FWA Detection	Indicates likely unbundling — billing separately for services that should be grouped under a single comprehensive code. Generated at a ~2% baseline rate. One of the most common forms of healthcare fraud.
claim_denial_flag	Binary	0 / 1	Financial / Revenue Cycle [NEW]	Indicates whether the insurance claim was rejected by the payer. Baseline denial rate ~10%, dramatically elevated when fraud indicators are present: upcoding adds 45 percentage points, duplicate claims add 55 points, unbundling adds 35 points. The fraud→denial correlation creates a learnable signal for denial-prediction models. Denial management is a top revenue cycle priority — denied claims average \$25 per appeal and 65% are never resubmitted.
denial_reason_code	String	CARC code	Financial / Revenue Cycle [NEW]	Claim Adjustment Reason Code (CARC) from the X12 standard, populated only when claim_denial_flag is 1. Codes are assigned intelligently based on

Column name	Type	Units / format	Field of interest	Description and purpose
				the underlying cause: duplicate fraud → CO-18, upcoding → CO-50/CO-11/CO-4, unbundling → CO-97. Clean denials draw from realistic CARC distribution led by CO-16 (lacks information), CO-197 (precertification absent), and CO-50 (not medically necessary). Used for denial categorization, appeal workflow routing, and root cause analysis.
equipment_type	String	Equipment name	Financial / Equipment Utilization [NEW]	Type of high-cost diagnostic or therapeutic equipment used during the encounter (MRI, CT Scanner, PET Scanner, Cath Lab, Echocardiogram, Ultrasound, X-Ray, Endoscope, Bronchoscope, Cystoscope, Dermatoscope). Equipment is mapped to diagnosis category for clinical realism: MSK cases use MRI/X-Ray/CT, CV cases use Cath Lab/Echo/CT, ONC cases use PET/MRI/CT. Used together with equipment_usage_minutes to compute the Equipment Utilization Rate KPI.
equipment_usage_minutes	Integer	Minutes	Financial / Equipment Utilization [NEW]	Number of minutes the equipment was operationally used for this encounter. Mean varies by equipment type: Cath Lab averages 60 minutes, MRI 45 minutes, PET 45 minutes, CT 25 minutes, Ultrasound 15 minutes, X-Ray 8 minutes. Used to compute Equipment Utilization Rate (used minutes ÷ available minutes) — a core capital asset productivity KPI for hospital finance teams.
consumable_supply_cost_usd	Float	USD	Financial / Cost Accounting [NEW]	Cost of general hospital consumables for this encounter (IV bags, syringes, gauze, gowns, dressings, catheters, wound care). Explicitly separate from kit_unit_cost_usd, which only tracks the surgical kit. Scales with length of stay (+\$35 per day) and severity. Mean ~\$256 per encounter, matching typical hospital consumable budgets. Used to compute the full Supply Cost per Admission KPI when combined with the surgical kit cost.

5. Human resources domain — staffing, burnout, and CAHPS

Twenty-one fields covering the workforce assigned to the encounter, the conditions under which they worked, their psychometric burnout state, and the resulting CAHPS patient experience scores.

Column name	Type	Units / format	Field of interest	Description and purpose
nurse_emp_id	String	NRS-XXXXX	HR / Identifier	Synthetic employee identifier for the primary attending nurse. Joins to the staff roster file for additional HR attributes. Used to compute nurse-level outcomes and turnover analytics.
nurse_role	String	Role	HR / Workforce	Nursing role category (RN, LPN, Charge RN). Drives the scope of practice and supervisory responsibilities recorded for the encounter.
nurse_unit	String	Unit name	HR / Assignment	Home unit of the nurse (which may differ from the patient's assigned unit if the nurse was floating). Used for unit-level vacancy and skill-mix analysis.
nurse_tenure_years	Float	Years	HR / Workforce	Years of employment at the institution. Lower tenure correlates with higher turnover risk and, U-shaped, with burnout. Used for retention modeling.
nurse_fte	Float	0.0–1.0	HR / Workforce	Full-time equivalent status of the nurse (1.0 = full-time, 0.5 = half-time). Used for productive-hours-per-patient-day calculations and budget modeling.
surgeon_emp_id	String	SRG-XXXXX	HR / Identifier	Synthetic employee identifier for the attending surgeon. Joins to the surgeon roster for specialty, NPI, and tenure attributes.
surgeon_specialty	String	Specialty	HR / Provider	Primary surgical specialty of the attending physician (General Surgery, Orthopedics, Cardiothoracic, Urology, Neurosurgery). Used for service-line productivity and outcomes reporting.
shift_hours	Integer	Hours	HR / Workforce	Length of the nursing shift during which the patient was cared for (8, 10, or 12 hours). Twelve-hour shifts are associated with higher fatigue and overtime accumulation.

Column name	Type	Units / format	Field of interest	Description and purpose
patients_per_nurse_ratio	Integer	Patients	HR / Staffing	Number of patients assigned to the attending nurse during this shift. The single most cited indicator in nurse staffing research. Higher ratios correlate with safety incidents, lower CAHPS scores, and higher burnout.
overtime_hours	Float	Hours	HR / Workforce	Overtime hours worked by the nurse during the relevant pay period. Used as an early-warning indicator for fatigue-related errors and as a leading indicator of impending turnover.
burnout_exhaustion_mbi	Float	0.0–6.0	HR / Wellbeing	Emotional Exhaustion subscale score from the Maslach Burnout Inventory (MBI), the gold-standard burnout instrument. Higher scores indicate greater emotional depletion. Drives downstream incident probability and CAHPS scores.
burnout_cynicism_mbi	Float	0.0–6.0	HR / Wellbeing	Depersonalization (Cynicism) subscale from the Maslach Burnout Inventory. Measures detachment and impersonal response toward patients. Correlated with the exhaustion subscale.
turnover_risk_index	Float	0.0–1.0	HR / Retention	Composite turnover risk score derived from burnout, tenure, and overtime hours. A leading indicator for nurse retention modeling and proactive intervention.
cahps_nurse_communication	Float	1.0–5.0	Patient Experience / CAHPS	Patient-reported score on how well nurses communicated. From the HCAHPS survey domain. Inversely correlated with nurse burnout and patient load.
cahps_doctor_communication	Float	1.0–5.0	Patient Experience / CAHPS	Patient-reported score on how well doctors communicated. From the HCAHPS survey domain. A core dimension of the CMS Star Rating program.
cahps_responsiveness	Float	1.0–5.0	Patient Experience / CAHPS	Patient-reported score on staff responsiveness — how quickly help was provided when called for. Strongly affected by patient-to-nurse ratios and unit occupancy.

Column name	Type	Units / format	Field of interest	Description and purpose
cahps_pain_management	Float	1.0–5.0	Patient Experience / CAHPS	Patient-reported score on pain management. Note that CMS removed this dimension from the Star Rating in 2018 but it is still tracked for internal quality improvement.
cahps_discharge_info	Float	1.0–5.0	Patient Experience / CAHPS	Patient-reported score on the quality of discharge information received. A key driver of post-discharge adherence and 30-day readmission risk.
cahps_care_transition	Float	1.0–5.0	Patient Experience / CAHPS	Patient-reported score on care transitions between providers and care settings. Added to HCAHPS in 2012; correlates with readmission rates.
cahps_cleanliness	Float	1.0–5.0	Patient Experience / CAHPS	Patient-reported score on the cleanliness of the hospital environment. A core HCAHPS dimension and a leading indicator of infection-control performance.
cahps_quietness	Float	1.0–5.0	Patient Experience / CAHPS	Patient-reported score on the quietness of the hospital environment, particularly at night. Correlates with sleep quality, healing, and overall patient satisfaction.

6. Patient experience and engagement — digital adoption and perception

Three new fields capturing patient-facing engagement metrics: telehealth and digital portal adoption, appointment scheduling lead time, and the gap between patient-perceived and actual wait times.

Column name	Type	Units / format	Field of interest	Description and purpose
telehealth_used_flag	Binary	0 / 1	Patient Experience / Digital Engagement [NEW]	Indicates whether the patient used digital tools for scheduling, consultation, or follow-up (patient portal, mobile app, video visit, e-check-in). Probability is age-correlated: starts at ~45% for patients under 40 and decreases by 0.5% per year of age beyond 40, modeling the well-documented digital divide. Used to compute the Telehealth and Digital Portal Adoption KPI, a core measure in HIMSS digital maturity assessments and CMS Promoting Interoperability scoring.
appointment_request_date	Date	YYYY-MM-DD	Patient Experience / Access [NEW]	Date the patient first requested the appointment that led to this encounter. Lead time varies by acuity: elective oncology and orthopedic procedures average 21 days, urgent cases (severity > 0.8) average 1 day, and routine cases average 7 days. Subtracting this from admit_timestamp yields the Appointment Scheduling Lead Time KPI — a core access-to-care measure tracked by the National Committee for Quality Assurance (NCQA) and a leading indicator of network adequacy.
patient_reported_wait_time_mins	Integer	Minutes	Patient Experience / Perception [NEW]	Patient-reported (perceived) wait time in minutes, captured via post-discharge survey. Computed by inflating the actual bed wait time by a factor of 1.2–1.8x with additional inflation when CAHPS responsiveness scores are low. Mean

Column name	Type	Units / format	Field of interest	Description and purpose
				perception ratio is approximately 1.48x actual wait — patients perceive their wait as roughly 50% longer than it actually was, matching established healthcare survey research. Used to compute the Perceived vs. Actual Wait Time gap KPI, a critical input for patient experience improvement programs.

7. Supply chain domain — surgical kits, inventory, and demand

Eleven fields covering the surgical supply kits used during the procedure, including current stock levels, reorder thresholds, lead times, and predictive demand forecasts.

Column name	Type	Units / format	Field of interest	Description and purpose
<code>surgical_kit_id</code>	String	SKU-XXXX	Supply Chain / Identifier	Stock keeping unit identifier for the surgical kit used in the procedure. Primary key for joining to the materials management system.
<code>kit_name</code>	String	Kit code	Supply Chain / Item	Functional name of the surgical kit (e.g., LAP-CHOLE-KIT, TKA-KIT, CABG-KIT). Each kit is mapped to one or more CPT codes through the procedure-to-supply linkage.
<code>kit_unit_cost_usd</code>	Float	USD	Supply Chain / Cost	Unit cost of the surgical kit. Drives the supply portion of the submitted charge and is a key input for procedure-level margin analysis. Note: this field tracks only the surgical kit; general consumables are tracked separately in <code>consumable_supply_cost_usd</code> .
<code>kit_current_stock</code>	Integer	Units	Supply Chain / Inventory	Current on-hand stock of the surgical kit at the time of the procedure. Used in conjunction with reorder point and weekly demand to calculate stockout risk.
<code>kit_reorder_point</code>	Integer	Units	Supply Chain / Inventory	Inventory level at which a replenishment order should be triggered. Computed as (lead time × average daily demand) plus a safety stock buffer.
<code>kit_lead_time_days</code>	Integer	Days	Supply Chain / Procurement	Number of days from order placement to delivery. Critical input for safety-stock calculations and supplier performance tracking.
<code>kit_expiration_date</code>	Date	YYYY-MM-DD	Supply Chain / Inventory	Expiration date of the kit lot in current stock. Drives FEFO (First-Expired-First-Out) issuing logic and waste reduction efforts.
<code>stockout_risk_flag</code>	Binary	0 / 1	Supply Chain / Risk	Indicates whether current stock has fallen at or below the reorder point. A leading operational indicator for procurement teams and surgical scheduling.
<code>weekly_procedure_volume</code>	Integer	Procedures	Supply Chain / Demand	Average number of procedures per week using this kit. Forms the basis

Column name	Type	Units / format	Field of interest	Description and purpose
				of demand forecasting and predictive procurement.
<code>projected_demand_4wk</code>	Integer	Units	Supply Chain / Demand Planning	Forecasted kit consumption over the next four weeks, computed from historical weekly volume. Used in just-in-time procurement and budget planning.
<code>days_of_supply</code>	Float	Days	Supply Chain / Inventory	Number of days the current stock is expected to last given recent demand. The single most actionable indicator for inventory managers and a standard supply chain KPI.

8. Quality and standards benchmarking

Four fields capturing alignment with international quality standards and regulatory benchmarks: HEDIS, ISO 7101 PDSA, and HIMSS EMRAM maturity.

Column name	Type	Units / format	Field of interest	Description and purpose
<code>hedis_hba1c_tested</code>	Binary	0 / 1	Quality / HEDIS 2025	Indicates that an HbA1c test was performed for a diabetic patient. The numerator for the HEDIS Comprehensive Diabetes Care (CDC) measure. Hospitals are expected to achieve close to 100% screening compliance for the diabetic denominator.
<code>hedis_hba1c_poor_control</code>	Binary	0 / 1	Quality / HEDIS 2025	Indicates that the diabetic patient has poor glycemic control, defined as HbA1c >9.0%. This is a HEDIS care gap that triggers payer-driven outreach and care management programs. Lower rates are better.
<code>pdsa_cycle_id</code>	String	PDSA-XXXXX	Quality / ISO 7101	Identifier of the Plan-Do-Study-Act improvement cycle linked to a safety incident, when one occurred. Provides traceability from incidents to documented quality improvement activity, as required by ISO 7101 healthcare quality management standards.
<code>himss_emram_stage</code>	Integer	0–7	Digital Maturity / HIMSS	HIMSS Electronic Medical Record Adoption Model (EMRAM) stage at the time of the encounter. Stages range from 0 (no EMR) to 7 (advanced analytics, full interoperability). Used as a covariate when comparing quality outcomes across digitally mature versus less mature settings.

Appendix A: Summary of new columns in v2.0

The following 11 columns were added in version 2.0 to extend KPI coverage. They are highlighted with light orange row shading throughout the document and tagged with [NEW] in the field-of-interest column.

Clinical (2 new)

- `hai_flag` — Hospital-Acquired Infection indicator. Drives the HAI Incidence KPI.
- `follow_up_within_48h_flag` — Post-discharge follow-up compliance indicator. Drives the Post-Discharge Follow-Up Compliance KPI.

Operations (1 new)

- `physician_assessment_timestamp` — Door-to-doctor moment. Combined with `triage_timestamp`, drives the ER Throughput and Wait Times KPI.

Finance (5 new)

- `claim_denial_flag` — Indicates payer rejection. Drives the Insurance Claim Denial Rate KPI.
- `denial_reason_code` — CARC code identifying the denial cause. Completes the denial KPI.
- `equipment_type` — High-cost equipment identifier (MRI, CT, Cath Lab, etc.).
- `equipment_usage_minutes` — Operational minutes per encounter. Combined with `equipment_type`, drives the Equipment Utilization Rate KPI.
- `consumable_supply_cost_usd` — General consumables (separate from surgical kit). Completes the Supply Cost per Admission KPI.

Patient experience (3 new)

- `telehealth_used_flag` — Digital tool adoption. Drives the Telehealth and Digital Portal Adoption KPI.
- `appointment_request_date` — Scheduling anchor. Combined with `admit_timestamp`, drives the Appointment Scheduling Lead Time KPI.
- `patient_reported_wait_time_mins` — Survey-derived perceived wait. Combined with the actual wait interval, drives the Perceived vs. Actual Wait Time KPI.

Appendix B: Standards and terminologies referenced

The following external standards and terminologies are referenced throughout this dataset. All synthetic data elements have been mapped to these standards to support interoperability and benchmarking.

Clinical terminologies

- ICD-10-CM (International Classification of Diseases, 10th Revision, Clinical Modification, FY2025) — diagnosis coding maintained by CMS and the National Center for Health Statistics.
- CPT (Current Procedural Terminology) — procedure coding maintained by the American Medical Association.
- HCPCS (Healthcare Common Procedure Coding System) — supplemental procedure coding maintained by CMS.
- LOINC (Logical Observation Identifiers Names and Codes) — laboratory observation coding maintained by the Regenstrief Institute.
- RxNorm — normalized medication terminology maintained by the U.S. National Library of Medicine.
- CARC (Claim Adjustment Reason Codes) — X12 standard codes used to communicate why a claim or service line was paid differently than it was billed.

Interoperability standards

- HL7 FHIR R4 (Fast Healthcare Interoperability Resources, Release 4) — the modern standard for healthcare data exchange. Resources used in this dataset include Patient, Encounter, Condition, Observation, Procedure, Claim, and MedicationRequest.

Quality and benchmark frameworks

- HEDIS 2025 (Healthcare Effectiveness Data and Information Set) — the National Committee for Quality Assurance (NCQA) measure set used by more than 90% of U.S. health plans for quality reporting.
- CAHPS / HCAHPS (Consumer Assessment of Healthcare Providers and Systems) — patient experience survey instrument administered under CMS authority.
- ISO 7101 (Healthcare quality management systems) — the international standard for healthcare quality management, including PDSA improvement cycles.
- HIMSS EMRAM (Electronic Medical Record Adoption Model) — the eight-stage digital maturity framework from the Healthcare Information and Management Systems Society.
- Joint Commission International, 8th Edition — international hospital accreditation standards.
- Maslach Burnout Inventory (MBI) — the gold-standard psychometric instrument for occupational burnout.
- CMS HACRP (Hospital-Acquired Condition Reduction Program) — value-based purchasing program tied to hospital-acquired infections and conditions.
- CMS BPCI (Bundled Payments for Care Improvement) — episode-based payment model that reinforces post-discharge follow-up.

Privacy and de-identification

- HIPAA Privacy Rule, 45 CFR §164.514(b)(2) — Safe Harbor de-identification method requiring removal of 18 specific identifier categories.

Important notice

This dataset is synthetic and should never be used for actual clinical decision-making, billing, or regulatory submission. It is designed exclusively for analytics development, model training, education, and demonstration.